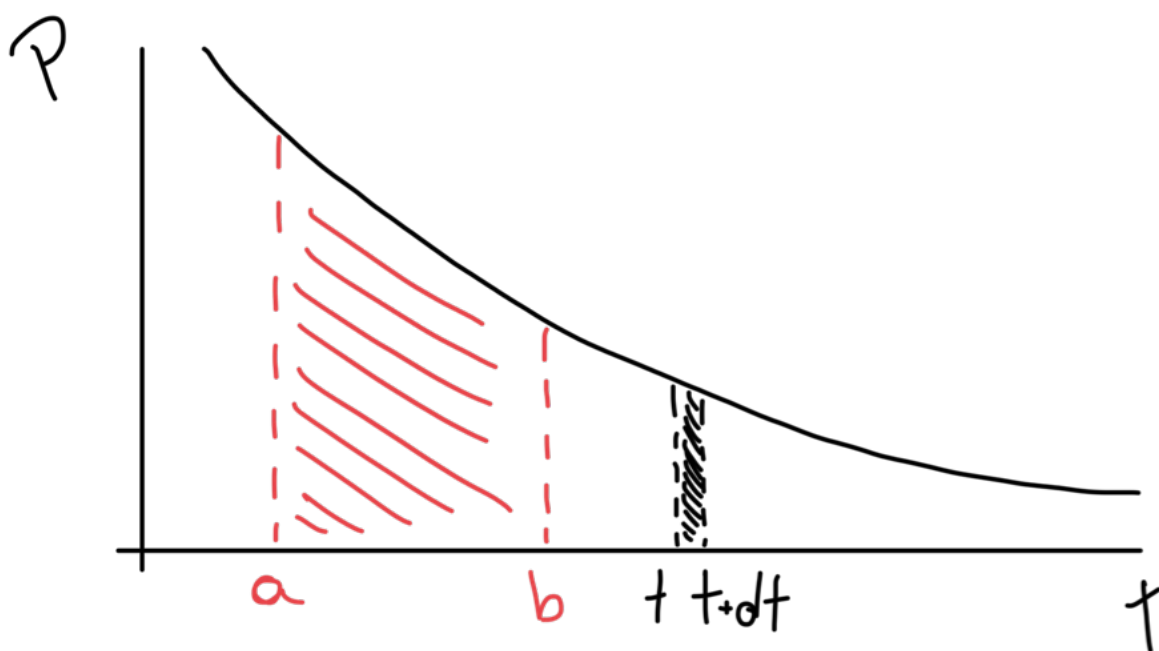


EXPONENTIAL DISTRIBUTION

A Market maker providing liquidity in AAPL options needs to know how long hold risk before another institutional client "clears" their position.

Def: A random time $T \sim \text{exponential}(\lambda)$

$$P(T=t) = \begin{cases} \lambda e^{-\lambda t} & t \geq 0 \\ 0 & t < 0 \end{cases}$$



$$P(t \leq T \leq t+dt) = \lambda e^{-\lambda t} dt$$

$$\begin{aligned} P(a \leq T \leq b) &= \int_a^b \lambda e^{-\lambda t} dt \\ &= \left[-e^{-\lambda t} \right]_a^b \\ &= e^{-\lambda a} - e^{-\lambda b} \end{aligned}$$

1. Exercise: Derive the CDF $F(t) = \int_{-\infty}^t P(t) dt$

$$\begin{aligned} F(t) &= \int_0^t P(t) dt = \int_0^t \lambda e^{-\lambda t} dt = \left[-e^{-\lambda t} \right]_0^t \\ &= (-e^{-\lambda t}) - (-e^0) = 1 - e^{-\lambda t} \end{aligned}$$

2. Exercise: Verify that $P(t)$ is a PDF

$P(t) = \lambda e^{-\lambda t}$ is PDF if:

① Non NEGATIVITY: $P(t) \geq 0 \quad \forall t$

→ λ is strictly positive $\lambda > 0$

→ $e^{-\lambda t}$ is always positive

SATISFIED ✓

② TOTAL PROBABILITY IS EQUAL TO 1

$$\int_0^{\infty} \lambda e^{-\lambda t} dt = 1$$

$$\lim_{b \rightarrow \infty} \int_0^b \lambda e^{-\lambda t} dt = \left[-e^{-\lambda t} \right]_0^b = 1 - e^{-\lambda \cdot \infty} = 1$$

SATISFIED ✓

$P(t)$ is a PDF ✓